Supplementary Materials for:

"Dual-Attention Deep Gaussian Processes for Multi-Objective Robust Parameter Design"

A Derivation of the Variational ELBO for Sparse GPs

This appendix provides a detailed derivation of the Evidence Lower Bound (ELBO) (Equation 4) described in Section 2.2. We consider the augmented GP model by introducing M inducing points $\mathbf{Z}$ and corresponding inducing variables $\mathbf{u}$. The joint probability of the observations $\mathbf{y}$, latent functions $\mathbf{f}$, and inducing variables $\mathbf{u}$ is given by:

$$p(\mathbf{y}, \mathbf{f}, \mathbf{u}) = p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{u})p(\mathbf{u}), \tag{A.1}$$

where $p(\mathbf{u}) = \mathcal{N}(\mathbf{0}, \mathbf{K_{uu}})$ is the prior over inducing variables, and $p(\mathbf{f}|\mathbf{u})$ follows the standard GP conditional distribution. To approximate the intractable posterior $p(\mathbf{f}, \mathbf{u}|\mathbf{y})$, we propose a variational distribution $q(\mathbf{f}, \mathbf{u})$. Following the framework of Titsias (2009), we enforce the variational posterior to factorize as:

$$q(\mathbf{f}, \mathbf{u}) = p(\mathbf{f}|\mathbf{u})q(\mathbf{u}), \tag{A.2}$$

where $q(\mathbf{u}) = \mathcal{N}(\mathbf{m}, \mathbf{S})$ is a free variational distribution to be optimized, and $p(\mathbf{f}|\mathbf{u})$ is fixed to the model prior conditionals. This corresponds to Equation (4) in the main text.The goal of variational inference is to minimize the Kullback-Leibler (KL) divergence between the approximate posterior and the true posterior, which is equivalent to maximizing the ELBO:

$$\mathcal{L} = \iint q(\mathbf{f}, \mathbf{u}) \log \frac{p(\mathbf{y}, \mathbf{f}, \mathbf{u})}{q(\mathbf{f}, \mathbf{u})} \, d\mathbf{f} d\mathbf{u} \tag{A.3}$$

Substituting (A.1) and (A.2) into (A.3), the ELBO can be expanded as:

$$\begin{aligned} \mathcal{L} &= \iint p(\mathbf{f}|\mathbf{u})q(\mathbf{u}) \log \frac{p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{u})p(\mathbf{u})}{p(\mathbf{f}|\mathbf{u})q(\mathbf{u})} \, d\mathbf{f} d\mathbf{u} \\ &= \iint p(\mathbf{f}|\mathbf{u})q(\mathbf{u}) \left[\log p(\mathbf{y}|\mathbf{f}) + \log \frac{p(\mathbf{u})}{q(\mathbf{u})} \right] \, d\mathbf{f} d\mathbf{u} \end{aligned}. \tag{A.4}$$

The term $p(\mathbf{f}|\mathbf{u})$ cancels out in the logarithm ratio, significantly simplifying the computation. The equation splits into two components:

(1) Expected Log-Likelihood:

$$\iint q(\mathbf{u})\, p(\mathbf{f} \mid \mathbf{u}) \, \log p(\mathbf{y} \mid \mathbf{f}) \, d\mathbf{f} \, d\mathbf{u} = \mathbb{E}_{q(\mathbf{f})}[\log p(\mathbf{y} \mid \mathbf{f})] = \sum_{i=1}^{N} \mathbb{E}_{q(f_i)}[\log p(y_i \mid f_i)] \,, \tag{A.5}$$

where $q(\mathbf{f}) = \int p(\mathbf{f}|\mathbf{u})q(\mathbf{u})d\mathbf{u}$ represents the marginal variational distribution.

(2) KL Divergence:

$$\int q(\mathbf{u}) \left[\int p(\mathbf{f} \mid \mathbf{u})\, d\mathbf{f}\right] \log \frac{p(\mathbf{u})}{q(\mathbf{u})}\, d\mathbf{u} = -\mathrm{KL}[q(\mathbf{u}) \,\|\, p(\mathbf{u})]\,. \tag{A.6}$$

Note that $\int p(\mathbf{f}|\mathbf{u})d\mathbf{f} = 1$. Combining (A.5) and (A.6), we obtain the final ELBO formulation as presented in Equation (4):

$$\mathcal{L}_{\mathrm{SGP}} = \sum_{i=1}^{N} \mathbb{E}_{q(f_i)}[\log p(y_i \mid f_i)] - \mathrm{KL}[q(\mathbf{u}) \,\|\, p(\mathbf{u})]\,. \tag{A.7}$$

B Proof of Unbiased Gradient Estimation via Reparameterization Trick

This appendix provides the mathematical justification for the gradient computation in the Doubly Stochastic Variational Inference (DSVI) framework, specifically concerning Equations (9)-(12) in Section 2.3. We aim to compute the gradient of the expected log-likelihood term in the ELBO with respect to the variational parameters $\boldsymbol{\phi}$ (i.e., $\mathbf{m}, \mathbf{S}$). The objective function is:

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi(\mathbf{f})}[\log p(\mathbf{y} \mid \mathbf{f})] = \int q_\phi(\mathbf{f})\, \log p(\mathbf{y} \mid \mathbf{f})\, d\mathbf{f}. \tag{B.1}$$

Directly differentiating (B.1) is problematic because the integration domain depends on $\boldsymbol{\phi}$. To resolve this, we apply the reparameterization trick defined in Equation (11):

$$\mathbf{f} = g(\phi, \boldsymbol{\epsilon}) = \boldsymbol{\mu}_\phi + \boldsymbol{\Sigma}_\phi^{1/2} \odot \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon}) = \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{B.2}$$

Using the Law of the Unconscious Statistician (LOTUS), the expectation over $q_{\boldsymbol{\phi}}(\mathbf{f})$ can be rewritten as an expectation over the auxiliary noise distribution $p(\boldsymbol{\epsilon})$, which is independent of $\boldsymbol{\phi}$:

$$\mathcal{L}(\phi) = \mathbb{E}_{p(\boldsymbol{\epsilon})}[\log p(\mathbf{y} \mid g(\phi, \boldsymbol{\epsilon}))]\,. \tag{B.3}$$

Now, the gradient operator $\nabla_{\boldsymbol{\phi}}$ can be moved inside the expectation:

$$\nabla_\phi \mathcal{L}(\phi) = \nabla_\phi \int p(\boldsymbol{\epsilon})\, \log p(\mathbf{y} \mid g(\phi, \boldsymbol{\epsilon}))\; d\boldsymbol{\epsilon} = \mathbb{E}_{p(\boldsymbol{\epsilon})}[\nabla_\phi \log p(\mathbf{y} \mid g(\phi, \boldsymbol{\epsilon}))]\,. \tag{B.4}$$

This expectation can be efficiently approximated using Monte Carlo sampling. For S samples of $\boldsymbol{\epsilon}_s \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\nabla_\phi \mathcal{L}(\phi) \approx \frac{1}{S} \sum_{s=1}^{S} \nabla_\phi \log p(\mathbf{y} \mid g(\phi, \boldsymbol{\epsilon}_s))\,. \tag{B.5}$$

Since $g(\boldsymbol{\phi}, \boldsymbol{\epsilon})$ is a deterministic differentiable function, the gradients can be computed via standard backpropagation (chain rule). This proves that the reparameterization trick provides an unbiased low-variance gradient estimator for optimizing the ELBO in Equation (9).

C Theoretical Justification of the Weighted ELBO

This appendix provides the theoretical basis for the sample-weighted ELBO objective function defined in Equation (16).Standard variational inference minimizes the KL divergence between the variational posterior $q(\mathbf{u})$ and the true Bayesian posterior $p(\mathbf{u}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{u})p(\mathbf{u})$. This is equivalent to maximizing the standard ELBO:

$$\mathcal{L} = \mathbb{E}_q[\log p(\mathbf{y} \mid \mathbf{f})] - \mathrm{KL}[q(\mathbf{u}) \,\|\, p(\mathbf{u})] \,. \tag{C.1}$$

To incorporate the sample-level attention mechanism rigorously, we adopt the framework of generalized variational inference (Knoblauch et al., 2019). In GVI, the standard Bayesian inference is generalized as an optimization problem involving a loss function ℓ, a divergence D, and a prior π:

$$q^* = \arg\min_q \left\{\mathbb{E}_q[\ell(\mathbf{y}, \mathbf{f})] + D(q \,\|\, \pi)\right\} . \tag{C.2}$$

In our DA-DGP framework, we define a weighted log-likelihood loss to emphasize the ROI:

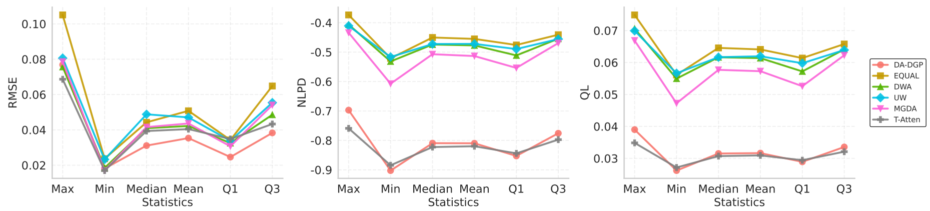
$$\ell_{\text{weighted}}(\mathbf{y}, \mathbf{f}) = -\sum_{k=1}^{p}\sum_{i=1}^{N} \tilde{s}_{i,k} \, \log p\big(y_{i,k} \mid f_{i,k}^L\big) \,, \tag{C.3}$$

where $\tilde{s}_{i,k}$ are the normalized attention weights derived in Equation (14). Setting the divergence D to be the KL divergence, the optimization objective becomes:

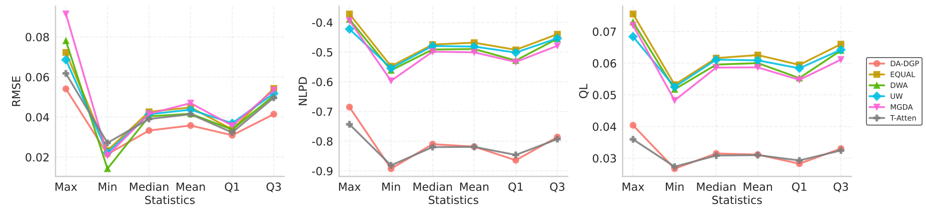
$$\mathcal{L}_{\text{DA-DGP}} = \sum_{k=1}^{p}\sum_{i=1}^{N} \tilde{s}_{i,k} \, \mathbb{E}_{q(f_{i,k}^L)}\left[\log p\big(y_{i,k} \mid f_{i,k}^L\big)\right] - \mathrm{KL}[q(\mathbf{u}) \,\|\, p(\mathbf{u})] \,. \tag{C.4}$$

This derivation confirms that our weighted objective (Equation 16) is not merely a heuristic modification but a valid GVI posterior approximation that minimizes a preference-weighted risk bound.

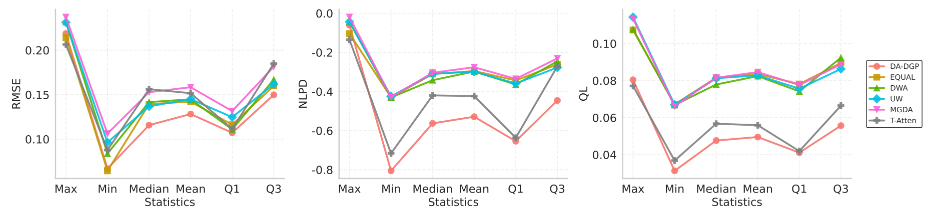
D Supplementary Visualizations of Experimental Results



(a) Task 1 (y_1)



(b) Task 2 (y_2)



(c) Task 3 (y_3)

Figure D.1: Performance metric curves for the three tasks.

E Sensitivity Analyses for the Numerical Example

This appendix reports the detailed sensitivity analyses referenced in Section 4.1 of the main paper. We consider three factors: the Gaussian bandwidth used in sample-level attention, the surrogate depth/hidden complexity, and the available training sample size. All statistics are aggregated over 20 independent runs, and the figures below visualize taskwise error bars for RMSE, NLPD, and QL.

Section 4.1 Gaussian bandwidth / 20
RMSE NLPD QL

E.1 Bandwidth Sensitivity

The bandwidth study confirms that moderate kernels are preferable. Averaged over three tasks, $\sigma = 0.5$ yields the best QL (**0.034774**), while $\sigma = 0.3$ attains the best RMSE (**0.069403**) and remains very close to $\sigma = 0.5$. In contrast, $\sigma = 0.1$ degrades both RMSE and QL, indicating over-localization of the training signal. Hence, the default choice $\sigma = 0.5$ provides the most balanced overall behavior.

$\sigma = 0.5$ QL **0.034774** $\sigma = 0.3$ RMSE **0.069403** $\sigma = 0.5$
$\sigma = 0.1$ RMSE QL $\sigma = 0.5$

Table E.1: Average sensitivity summary for Gaussian bandwidth.

Gaussian

Condition	Avg. RMSE	Avg. NLPD	Avg. QL
$\sigma = 0.1$	0.090653	-0.611964	0.044973
$\sigma = 0.3$	**0.069403**	-0.734488	0.035847
$\sigma = 0.5$	0.070831	**-0.750523**	**0.034774**
$\sigma = 0.7$	0.075690	-0.737188	0.035506
$\sigma = 1.0$	0.078762	-0.731904	0.035760
$\sigma = 1.5$	0.082363	-0.717340	0.036645

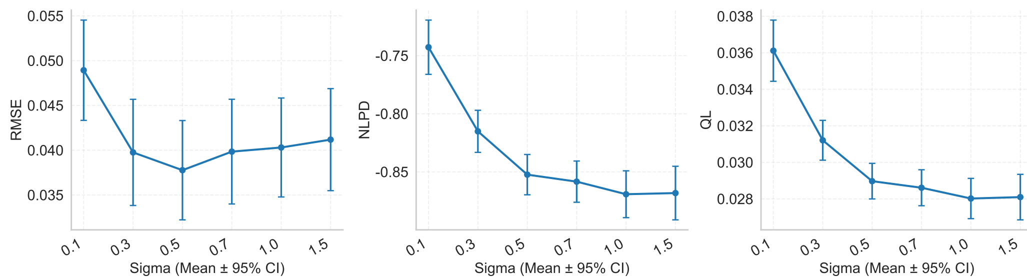
E.2 Depth and Hidden Complexity Sensitivity

The depth study shows that the default configuration is not only simpler but also clearly superior. The 2L-D5 model achieves the best average RMSE (**0.070808**), NLPD (**-0.751518**), and QL (**0.034732**). Shallower 2L-D2 underfits, while deeper 3L-D5, 4L-D5, and 5L-D5 rapidly deteriorate, suggesting that the current 5-dimensional benchmark with 400 training samples does not benefit from additional depth.

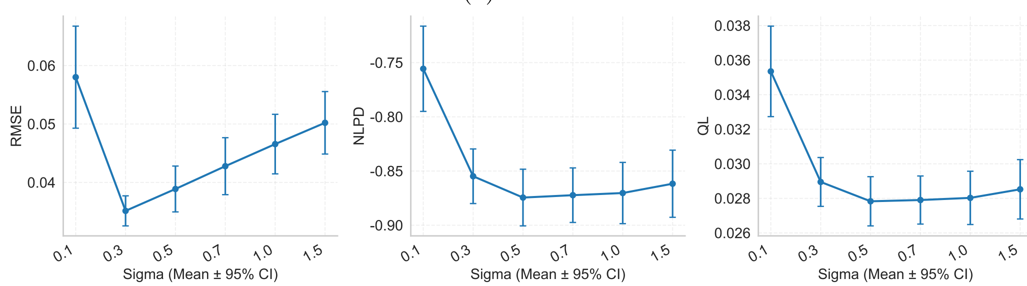
2L-D5 RMSE **0.070808** NLPD **-0.751518** QL **0.034732**
2L-D2 3L-D5 4L-D5 5L-D5 5 400

E.3 Sample Size Sensitivity

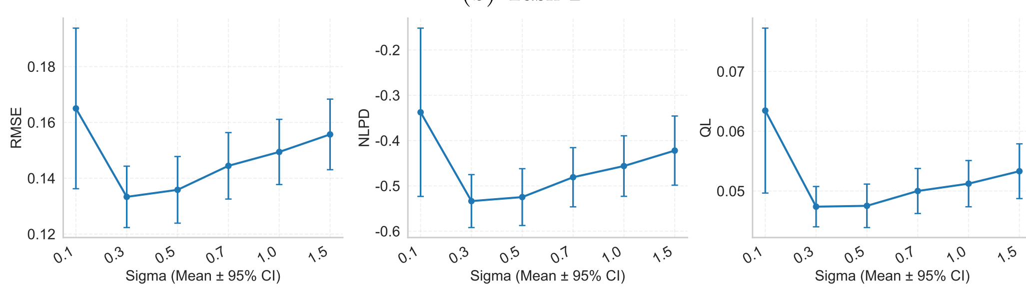
The sample-size experiment highlights a strong finite-sample effect. Performance improves monotonically as the training size grows from 20 to 400. In particular, the average QL



(a) Task 1



(b) Task 2



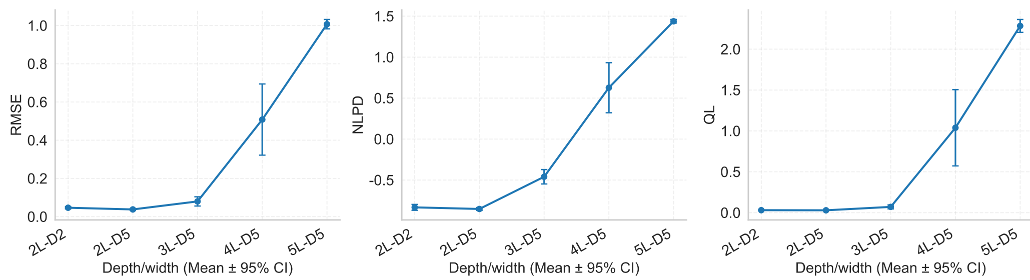
(c) Task 3

Figure E.2: Taskwise error-bar plots for Gaussian bandwidth sensitivity.

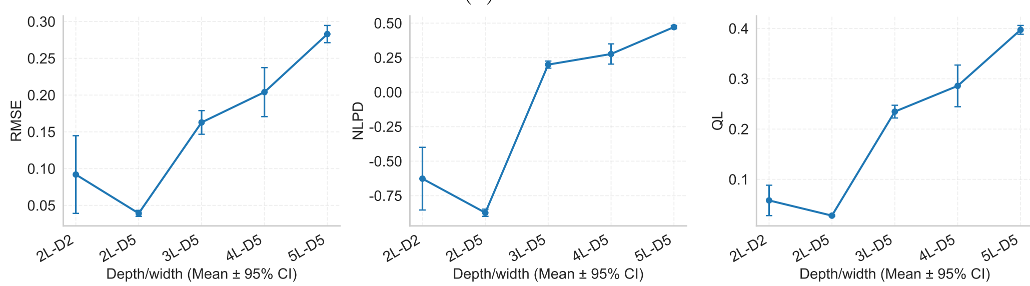
Gaussian

Table E.2: Average sensitivity summary for depth and hidden complexity.

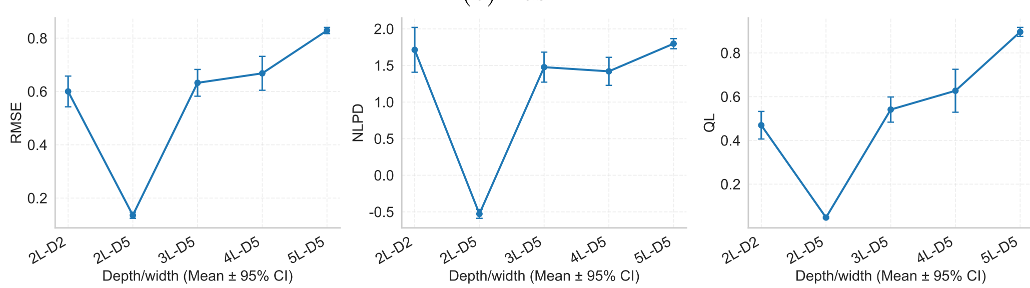
Condition	Avg. RMSE	Avg. NLPD	Avg. QL
2L-D2	0.246244	0.084328	0.185969
2L-D5	**0.070808**	**-0.751518**	**0.034732**
3L-D5	0.291672	0.405546	0.282074
4L-D5	0.460090	0.774311	0.650473
5L-D5	0.706363	1.235559	1.192501



(a) Task 1



(b) Task 2



(c) Task 3

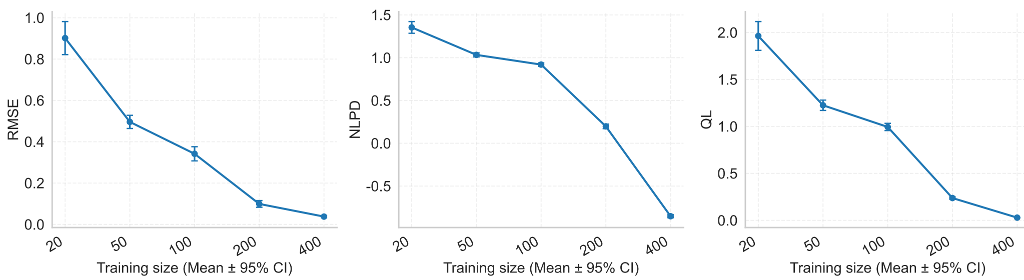
Figure E.3: Taskwise error-bar plots for depth and hidden-complexity sensitivity.

drops from 1.439658 at $n = 20$ to **0.034749** at $n = 400$, and the average RMSE drops from 0.592939 to **0.070777**. The pronounced gain from $n = 200$ to $n = 400$ further indicates that 400 samples are not excessive for this heterogeneous benchmark.

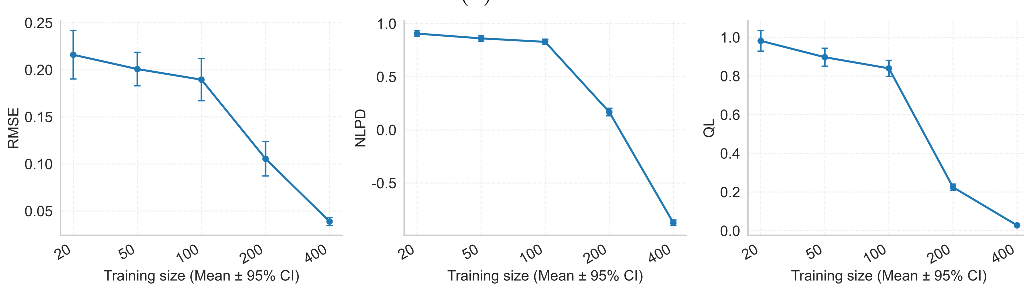
20 400 QL $n = 20$ 1.439658 $n = 400$ **0.034749**
RMSE 0.592939 **0.070777** $n = 200$ $n = 400$ 400

Table E.3: Average sensitivity summary for training sample size.

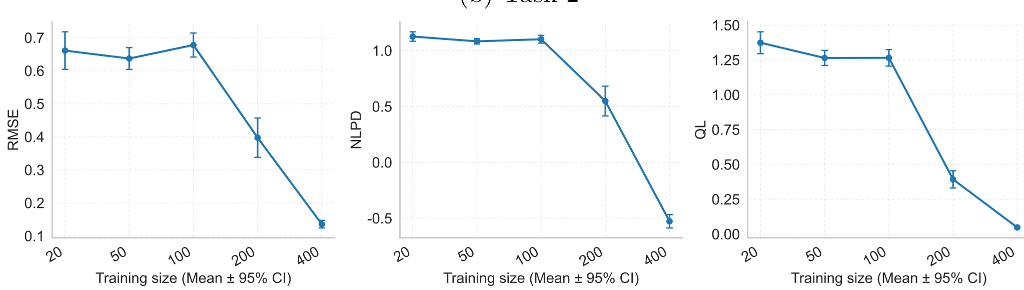
Condition	Avg. RMSE	Avg. NLPD	Avg. QL
$n = 20$	0.592939	1.128905	1.439658
$n = 50$	0.444696	0.992585	1.129003
$n = 100$	0.403108	0.950088	1.033113
$n = 200$	0.200731	0.304830	0.285129
$n = 400$	**0.070777**	**-0.751240**	**0.034749**



(a) Task 1



(b) Task 2



(c) Task 3

Figure E.4: Taskwise error-bar plots for sample-size sensitivity.

References

Titsias, M. (2009). Variational Learning of Inducing Variables in Sparse Gaussian Processes. *Artificial Intelligence and Statistics (AISTATS), PMLR*, 5, 567–574.

Knoblauch, J., Jewson, J., & Damoulas, T. (2019). Generalized Variational Inference: Three arguments for deriving new Posteriors. *arXiv preprint arXiv:1904.02063*.